

Sudheendra

Experience Summary
13+ years in Embedded Multimedia, Middleware, System Apps development in Linux and Android platform

Technical skills	
Programming Languages	• C, C++, Basic Java, JNI • TMS320C54x Assembly
DSP	• TMS320c54x, TMS320c67xx, • TI TDA2ex, OMAP5912
Operating System	• Android, Embedded Linux, Symbian OS • Windows XP, Linux / Unix,
CODECS	• JPEG, TIFF, MP3, MPEG2, MPEG4, H.263, H.264, MotionJpeg, SMV, G.729AB, Acoustic Echo Canceller
FRAMEWORK & TECHNOLOGY	• AMD MAI Multimedia Framework, • TI Vision SDK, TI OMAP Reference Framework, Doubango GAN, VoIP, GPRS Service • OpenCore Framework, AudioFlinger
PRODUCT	• Car DVR, Mobile Handset, Portable Media Player, STB • Digital Video Recorders
SKILLSET	• Android Framework, Middleware customization • Multimedia and streaming Framework and Application development • Android Application development • Embedded Linux Application and Middleware • Video and Audio CODEC development and integrations • System programming and IPC mechanisms • Socket and Client-Service programming
TOOLS	• Android Eclipse, Android Studio, Wireshark, • Visual Studio, Code Composer Studio 2.0/v2.1, • Metrowerks Code Warrior, Matlab 6.0 • Logical Analyser, Simulink, Tornado, Scilab 2.6, ScopeDSP, Pspice • Git, SVN

DOMAIN EXPERTISE / KEY ACTIVITIES
• Strong hands on experience in Multimedia products development and customization and technical leading, team building expertise • Requirements gathering and analysis • Strong experience in C++ and OOAD developments • Good experience in Embedded multimedia Frameworks, Stacks and Application development in Android and Embedded Linux • Experience in Android AOSP

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- Good experience in Embedded Linux platform
- Knowledge in Embedded Linux driver
- Good knowledge in socket and Client-Service programming
- Working experience in video streaming such as RTSP
- Strong knowledge in Video and Audio codecs and Integrations with the Frameworks
- Developed features related to audio like Echo cancellation
- Good experience in Audio and Video codec porting and optimization DSPs
- Tuning of AEC for Earpiece, Speaker, Wired Headset & Bluetooth modes of operation for various Android Smartphones comprising Samsung Galaxy Series, LG, Motorola, Huawei, etc
- Fixing of issues related to GAN Client for various handsets
- Identifying/Collating the innovative ideas and directing / leading the team in delivering the solution meeting customer set quality & timelines expectation.
- Tracking of the activities of the team members
- Identifying key risk areas in the project and report to the management
- Development of the feature for the Motorola DCT 64xx STBs
- Fair knowledge in Hardware debugging
- Integrating the feature with the System
- Unit testing of the feature
- Resolution of the bugs pertaining to the Audio, Video, Demux, Tuner, Graphics modules

Relevant Projects

Project	Intelligent Transport System (ITS) – For Indian Automotive OEM
Description	The scope of this project is to design and develop the Fully automotive grade Intelligent Transport System. This solution includes multiple sub-systems like Digital Video Recorder, Telematics, Power over Ethernet, Passenger Information System and Power control module. Below are the features of ITS system, • DVR ○ Support for Analog and IP camera recording ○ Live video streaming ○ Single and quad display mode ○ Reverse parking camera assistance ○ Multiple recording mode(auto, user, event, etc) ○ Data backup over Wi-Fi and USB3.0 • Telematics ○ AIS140 specific tracking system ○ Driver behavior monitoring ○ eCall support ○ Hotspot support ○ OTA Firmware upgrade • Power over Ethernet

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	○ Provision for PoE IP camera ○ System diagnostic connectivity • Passenger Information System ○ Routing management ○ Route announcement (audio and text) ○ Emergency announcements • Power Control Module ○ Automotive specific power management ○ Support for different power modes
Roles & Responsibilities	• Make build system to build and installation of ITS applications • Customized the TI's McFW framework to support simultaneous recording with RTSP streaming • IP Camera applications performances optimization • RTSP Server porting and integration • FFMPEG integration for recording • Interface development for HMI application. • Customer communication and release management • Platform software and ITS application integrations
Languages & Tools used	• C, TI TDA2Sx Vision SDK • Embedded Linux (Kernel V 4.4)

Project	3 Channel Car Video Recorder (CarDVR) - For Indian Automotive OEM
Description	The scope of this ongoing project is to design and develop the 3 Channel Car Digital Video Recorder for the Leading Indian Car Manufacture company for their new vehicle platform. This CarDVR product is built with TI's latest TDA2 SoC. The DVR has two analog cameras, one in the front of the vehicle and one in the back of the vehicle. One HD camera is placed inside the vehicle to record the occupants of the vehicle. The DVR platform supports live preview over USB through UVC (H.264) to a Head Unit of the vehicle. The recording of the 3 cameras is supported with various use cases. • The complete DVR hardware is being designed in house. • The DVR communicates with the Telematics Control Unit of the vehicle via Wifi. • The DVR is interfaced to the vehicle bus through CAN. • The CAN stack is running on the MCU which is in turn connected to the SoC of the DVR. • Camera boards were connected to the DVR board using the serialiser/deserialiser combination and video decoder.

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	Vision SDK from TI with linux kernel version 4.4 on A15 core is used for the DVR software development. BSP, Driver development, Recording Middleware and Application development, Entire system application are scope of this project.
Role and Responsibilities	- Handled the core recording module of the product - Added single channel video pipeline to capture (CMOS/Analog camera), encode and record as H.264 elementary stream - Developed new plugin to write the encoded data in shared memory with pre-event buffering feature - Developed Link APIs to integrate the circular buffer with UVC streaming and MP4 recorder - Developed test applications to validate video streaming/recording features - Enhanced the existing Make build system to support build and installation of DVR applications
Languages & Tools used	- C, TI TDA2ex Vision SDK, Embedded Linux - STM SPC560C40L1, Git

Project:	VoWi-Fi Calling Application Development/Porting/Defect Fixing for Android Phones
Description	The VoWiFi application enables a native Wi-Fi Calling experience on Android handsets. With this App, operators can quickly and significantly expand the availability of their VoWiFi service and bring the benefits of improved indoor coverage to many more new and existing subscribers. Once installed, the client operates in the background and subscribers continue to make calls using the native dialer, contacts app, call log or with any third party applications that initiate standard mobile calls.
Role and Responsibilities	- Requirements gathering and analysis - Developed the audio module to handle voice calls - Integration of the open source SIP stack in the middleware - Fixing of issues related to audio for various handsets like: Samsung Galaxy Series, LG, Intel, Motorola, Sharp, etc - Key role in solving the technical issues arising during providing solution to the customer. - Asses the priority of the tasks and allocation accordingly to the team members. - Tracking of the activities of the team members - Identifying key risk areas in the project and report to the management
Languages & Tools used	- C, C++, JAVA - Android NDK

Project:	Integration of Acoustic Echo Canceller into GanLite Client
Description	Smart Wi-Fi GanLite Client is based on the 3GPP UMA/GAN standard, turns existing Wi-Fi access points into seamless extensions of the mobile network. Using this client, subscribers can now resolve indoor mobile coverage problems by receiving their voice services over existing Wi-Fi access points. Typically, in case of a Cellular call the entire voice processing for a voice call is performed in handset modem stack. It is designed such that the Modem stack would be interfaced with an hardware AEC at the chipset level to cancel echo during the call. However, in case of WiFi call most part of the voice processing is performed in the middleware due to the reason that the voice needs to be propagated as RTP packets over TCP/IP stack. Thus
Role and Responsibilities	• Requirements gathering and analysis • Based on the requirements identify the interfaces of both the GAN & Echo Canceller • Communication between Canceller algorithm and Audio Flinger modules and customizing in AOSP • Implemented Recorder and Playback managers in middleware using the ALSA interface • Developed a thin layer between GAN and Echo Canceller to handle the operations involved in echo cancellation • Configuring of the tuning parameters for AEC
Languages & Tools used	• C, JAVA • Android NDK

Project:	Ello Application for Android phones
Description	This Ello App is a social networking application similar to WhatsApp, Viber, etc which has features like: sending chat messages, sending of multimedia contents - images/audio/video and voice call too. The sending and receiving of messages/multimedia contents is based on RCS protocol.
Role and Responsibilities	• Requirements gathering and analysis • Developed DTMF feature for the voice call service being provided in the Ello Application • Custom Recorder and Player Managers development • Fixing of issues related to audio for various handsets
Languages & Tools used	• C, JAVA • Android

Project:	Smart WiFi GanLite Client
Description	Smart Wi-Fi GanLite Client is based on the 3GPP UMA/GAN standard, turns existing Wi-Fi access points into seamless extensions of the mobile network. Using this client, subscribers can now resolve indoor mobile coverage problems by receiving their voice and data services over existing Wi-Fi access points. Smartphones have highlighted the coverage and capacity challenges of mobile networks, especially in indoor locations. Operators have started to look to Wi-Fi, already installed in many homes and offices, to address these issues.
Role and Responsibilities	• Requirements gathering and analysis • Developed features related to audio • Tuning of AEC for Earpiece, Speaker, Wired Headset & Bluetooth modes of operation for various Smartphones comprising Samsung Galaxy Series, LG, Motorola, Huawei, etc • Fixing of issues related to GAN Client for various handsets
Languages & Tools used	• C, C++, JAVA • Android NDK

Project:	Portable Media Player Product
Description	Portable Media Player is the multimedia device used for playback of Video, Still Image and Audio files. In addition the device supports: • Image manipulation – Rotate, Zoom • Image/Audio/Video files storage utilising SD/CF cards • Image printing by connecting this device with the PictBridge compatible printers • Image formats – JPEG, TIFF, Customer specific RAW formats • Audio formats – MP3, Wav, PCM, ADPCM • Video formats – MPEG1, MPEG2, MPEG4, MotionJpeg • USB 2.0 / SD/CF card hardware interface
Role and Responsibilities	• Product requirement gathering and analysis • Design of the end-to-end software solution for the media player • Development of image processing algorithms – Resizing, Rotation, Zoom, Thumbnails • Development of customer specific RAW image parsers • Development of Mpeg1/2/4, MJPEG Video file parsers • Development of MP3, Wav Audio file parsers

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	• Development of MP3 ID3 tag parser • Customizing of the Jpeg decoder interface • Optimization of the Jpeg decoder • Integration of the AMD MAI framework , Image / Audio / Video modules with the media player software system • Unit testing of each of the developed component • Audio/Video/Image module level testing • Integration Testing • Contributed in generating of Image / Audio / Video file samples required for boundary check / Negative scenarios • Contributed towards the Product User Guide
Languages & Tools used	• C, C++, Embedded Linux 2.6.11 • Linux cross-compiler tool chain compatible for AMD AU1200 Processor

Project:	Set-Top Box Product development
Description	The STB DCT 64XX development activities involved the SOC (Selectable Output Control) feature development and issues resolution. The SOC feature is developed to inhibit the end users from recording/copying the High Value Video Content over the STB Digital Video Output ports.
Roles & Responsibilities	• Design of SOC feature • Development of the feature for the Motorola DCT 64xx STBs • Integrating the feature with the System • Unit testing of the feature • Resolution of the bugs pertaining to the Audio, Video, Demux, Tuner, Graphics modules
Programming Language	• C, Vrtx • Metrowerks Code Warrior IDE, Rational Clearcase

Project:	Porting of TMS320C541 source code compatible G.729AB Speech CODEC on TMS320C6711 DSP
Description & Responsibilities	G.729AB is the standard for speech encoding and decoding done to compress the speech signals, confirmed to ITU-T standards. The "C" source code of G.729 CODEC compatible for TMS320C541 DSP is ported on TMS320C6711 DSP. This involved changing the data types and basic operations. The changes result in Overflow/underflow, which is taken care by saturating the values to either maximum or minimum respectively.
Role and Responsibilities	• High-level Design of porting procedure. • Review of the Reference Code (TMS320C541 compatible Source Code) • Porting of G.729 codec on TMS320C6711. • Used "C" Language

	Used the tool Code Composer Studio v2.1
Environment	Windows 98, C, Code Composer Studio v2.1

Project:18	MPEG-1 LAYER 3 (MP3) Audio decoder Confirming to ISO MPEG audio standards
Description & Responsibilities	The key to the efficiency of MP3 is perceptual coding – in other words ignoring those noises that are masked by more dominant sounds, and therefore not perceived by the ear. In order to do this the MP3 algorithm includes a psycho-acoustic model of the human ear. By ignoring information not perceived by the ear, the signal can be represented by fewer bits without loss of quality
Role and Responsibilities	- Involved in High Level Design of the decoder. - Implemented the Huffman variable length decoding method in High-level language. - Used 'C' as the High-level language. - Used various tools like VC++, Matlab 6.0, Code Composer Studio 2.0.
Environment	Windows 98/2000, VC++, Matlab 6.0, Code Composer Studio 2.0.

Project:	Motion Jpeg Decoder
Description & Responsibilities	Motion Jpeg is the video format used in Still Image cameras and Video cameras to store the recorded video. The encoded video is stored in the AVI file container as recommended by the Open DML AVI standard. The activity comprised of the development of the MJPEG video decoder. The decoder processes the encoded MJPEG video file and generates the raw output video in RGB / YUV formats.
Role and Responsibilities	- Involved in Design of MJPEG video decoder - Involved in the Development of MJPEG AVI parser - Involved in integration of the MJPEG AVI parser with the JPEG decoder - Involved in customizing the Input / Output interface of the MJPEG video decoder to cater to the Customer requirements - Involved in Unit testing of the MJPEG video decoder
Languages & Tools used	- C, Embedded Linux - Linux compiler, GDB, Valgrind, YUV Player

Project:	Porting and optimisation of H.263 video codec "C" source code on TI 55x DSP subsystem of OMAP 5912
Description & Responsibilities	H.263 is the video coding standard recommended by ITU-T for MMS video messaging and for 3G Mobile Video Telephony. The "C" source code of the H.263 code is ported on TMS320C55x DSP. The code is optimised for speed to

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	achieve real-time performance. The codec is integrated with the ARM application running on the ARM subsystem of the OMAP.
Role and Responsibilities	• High-level Design of porting H.263 on TMS320c55x • Porting of H.264 on TMS320c55x • High-level design for optimization • Optimisation of H.263 on TI 55x
Environment	• Windows 98, Linux, C, • Code Composer Studio v2.0, VC++ IDE

Project:	Porting of ANSI C compatible H.264 CODEC on TMS320C6711 DSP
Description & Responsibilities	H.264 is the video coding standard. It is the advanced video coding (AVC) standard also referred as MPEG-4 Part 10. The "C" source code of the CODEC is ported on TMS320C6711 DSP. The code is optimized for speed to achieve real-time performance.
Role and Responsibilities	• High-level Design of porting. • Porting of H.264 on TMS320c6711 • High-level design for optimization
Environment	• Windows 98, C, Code Composer Studio v2.1, • VC++ IDE

Project:	H.264/MPEG-2 Video Transcoding
Description & Responsibilities	The H.264 is expected to replace the use of Mpeg-2 video compression in digital video system. This creates an important need for transcoding technologies that transcode the widely available Mpeg-2 compressed video to H.264 compressed format and vice versa. The high transcoding complexity can be reduced by reusing the information gathered in the video decoding process.
Role and Responsibilities	• Study of Video Transcoding. • High-level Design of transcoding architecture
Environment	• Windows 98, C, Code Composer Studio v2.1, VC++ IDE

Project:	AT Command Software Development, Maintenance and Support
Project Description	This activity comprises of AT Commands feature development, CR fixing and Testing of the AT commands software module of 3G/GSM Mobile handsets. AT commands are through UART interface.
Roles & Responsibilities	• Requirements gathering and analysis • Based on the requirements that could be feature development / CR fixing / Test Cycles, estimate the efforts to deliver the same. • Asses the priority of the tasks and allocation accordingly to the team members.

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	• Planning for the Knowledge Capture of AT Command software, that would be outsourced to TechM premises in future. • Identifying/Collating the innovative ideas and directing / leading the team in delivering the solution meeting customer set quality & timelines expectation. • Tracking of the activities of the team members • Identifying key risk areas in the project and report to the management • Solving the technical issues arising during providing solution to the customer. • Managing the resources.
Languages & Tools used	• C, C++, Android • Logical analyzer for debugging the UART interfaces

Project:	Mobile Applications and data content reception Testing
Description & Responsibilities	The data contents such as images, audio, video and MIDP downloading and running/playing the same was tested on various 2.5G/3G handsets. The testing was done for the validation of the content delivery by the client's Content Delivery Server.
Role and Responsibilities	• Testing of validation of data contents for Nokia, Samsung, Sony Ericsson, Siemens • Generate the test/validation report. • Used the GPRS Data service • Used Mobile Handset hardware
Environment	• Symbian OS, GPRS Service

Project:	Smart antenna
Description & Responsibilities	Smart antenna is an efficient approach to increase the channel capacity of the wireless networks. The smart antenna system as a whole is the combination of multiple antenna elements with signal processing capabilities. The capacity of wireless network is limited due to the co-channel interference. The smart antenna system reduces the interference level, resulting in the increase of channel capacity
Role and Responsibilities	• Study of smart antenna system • High Level Design of Smart Antenna System Proof of Concept • Modeling of the System. • Used the tool, Matlab
Environment	• Windows 98, Matlab

Project:	XDAIS compliance for the Selectable Mode Vocoder
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Description & Responsibilities	The eXpress DSP initiative by Texas Instruments has set standard rules and guidelines for the DSP algorithm development. This followed, allows system integrators to quickly assemble production-quality system from one or more such algorithms. Thus, this standard is intended to enable a rich Commercial-off-the-shelf (COTS) marketplace for DSP algorithm technology and to significantly reduce the time-to-market for new DSP-based products. The Selectable Mode Vocoder is made compliant to eXpressDSP standard.
Role and Responsibilities	• Generating framework compliant to XDAIS • Modify the SMV source code to comply with XDIAS framework. • Integrate the modified SMV code with the XDAIS Framework. • Testing and validation of the SMV codec for XDAIS compliance. • Getting the XDAIS certification from Texas Instruments. • Used 'C' as the High level Language • Used various tools like Code Composer Studio, eXpressDSP tools (QualiTI)
Environment	• Windows 98, C, Code Composer Studio v2.1

Project:	Implementation of video decoder conforming to ISO/IEC 14496-2 Standards
Description & Responsibilities	ISO/IEC 14496 is a standard for interactive multimedia applications commonly known as MPEG-4. The project involves development of video codec conforming to ISO/IEC 14496-2.
Role and Responsibilities	• High Level Design of the mpeg-4 video decoder for simple profile. • Implemented the Inverse Discrete Cosine Transform in High-level language. • Testing of the same. • Used 'C' as the High level Language • Used various tools like VC++, Matlab 6.0
Environment	• Windows 98

Project:	Image Data Compression based on Baseline JPEG Standard
Description & Responsibilities	Image Data Compression refers to the reduction in the amount of data required to represent a digital image. The main stages involved in the Encoder part of the image compression method are DCT, Quantization, Zigzag scan, DPCM on DC components, Run length coding and Huffman Coding. The reverse process is followed in the decoder part of the compression method.
Role and Responsibilities	• Software development of the JPEG decoder in C
Environment	• Windows 98/2000, VC++, • Matlab 6.0, Code Composer Studio 2.0

Project:	Implementation of JPEG2000 encoder
Description & Responsibilities	The JPEG 2000 Standard is formulated for still image data compression. The encoder involves Preprocessing (tiling concept), Discrete Wavelet Transform, and Quantization, Bit plane coding, Arithmetic coding and Code stream organization. The reverse process is carried out in the decoder part of the CODEC
Role and Responsibilities	• Software development of the JPEG decoder in C
Environment	• Windows 98/2000, VC++, Matlab 6.0, • Code Composer Studio 2.0.